STARTING AND RECHARING

STARTING AND RECHARGING - DESCRIPTION

The ignition and recharging circuit comprises the battery, starter motor and alternator.

The battery (12V) is low maintenance type.

The starter motor consists of a d.c. motor supplied by the battery and an excitation electromagnet.

When the ignition key is turned as far as it will go (AVV), the motor windings are supplied to generate the electromagnetic forces that are used to turn the starter motor pinion. This simultaneously activates the electromagnet that operates the mechanism that causes the pinion to mesh with the flywheel ring gear and thus turn the crankshaft.

The alternator recharges the battery during normal engine rotation.

The alternator shaft (rotor) is turned by the crankshaft via a belt. When supplied by an excitation current, the rotor generates a magnetic field that sets up an alternating current in the fixed winding (stator).

A diode rectifier bridge at the back of the alternator allows the alternating current to be transformed into a direct current that is sent to recharge the battery.

A voltage regulator built into the alternator maintains the power supply at a constant voltage (around 14 V) throughout all load variation and engine speed ranges.

Recharging system efficiency is controlled by the Body Computer, which measures the D+ signal from the alternator with the engine running.

If the voltage measured is insufficient, the warning light in the instrument panel is switched on as well as the display of a message.

As an option, an upgraded potentiometer (160A) that is designed for connection to oversized cable cross-sections may be fitted.

STARTING AND RECHARGING - FUNCTIONAL DESCRIPTION

The Body Computer M001 receives a direct battery power supply at pin 1 (protected by fuse F38 of B002) and 18 of the coupling with the junction unit under the dashboard B002; it also receives an ignition-controlled power supply (INT) at pin 9 of the same coupling.

The M001 Body Computer is also connected to the C022 central dashboard earth via pins 10 and 19 of connector B and via pin 20 of the junction with the B002 junction unit under the dashboard (output from pin 10 of connector B of the B002 junction unit).

On turning the key in the H001 ignition switch to the end position (AVV), the starter motor A020 (connector C) electromagnets winding receive a power supply from pin 3 of connector A.

This direct battery power supply reaches the H001 ignition switch (pin 2 of connector A) via a line protected by fuses F70 of the B099 fuse control unit and F03 of the B001 engine compartment junction unit.

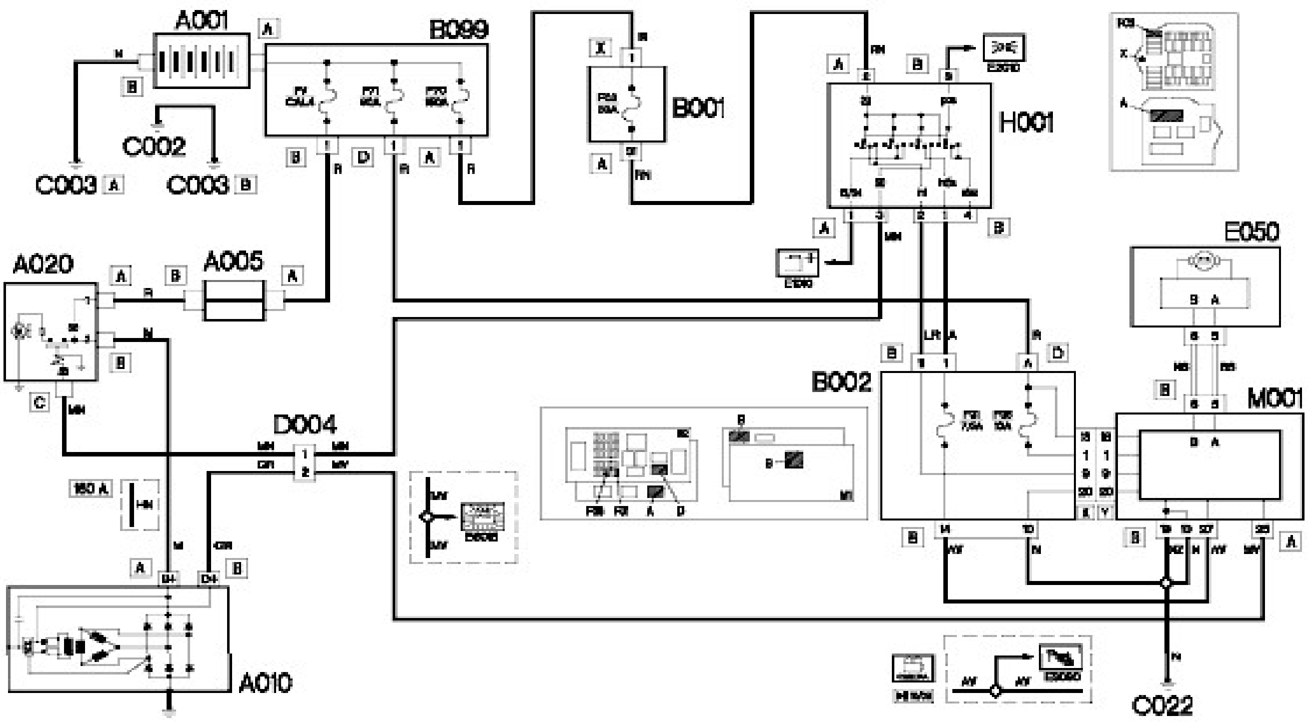
Connector A for the starter motor A020 receives a power supply directly from the battery via the line protected by the dedicated "powerval" fuse (CAL4) of the fuse box on the battery B099 and the shunt fuse for terminal board A005.

The direct current generated by the A010 alternator is sent from connector A (B+) to connector B of the A020 starter motor, which is internally connected to the line connected to the A001 battery.

The connection between connector B (D+) of the A010 alternator and pin 25 of connector A of the M001 Body Computer allows the performance of the alternator itself to be diagnosed (in the event of an insufficient recharging level).

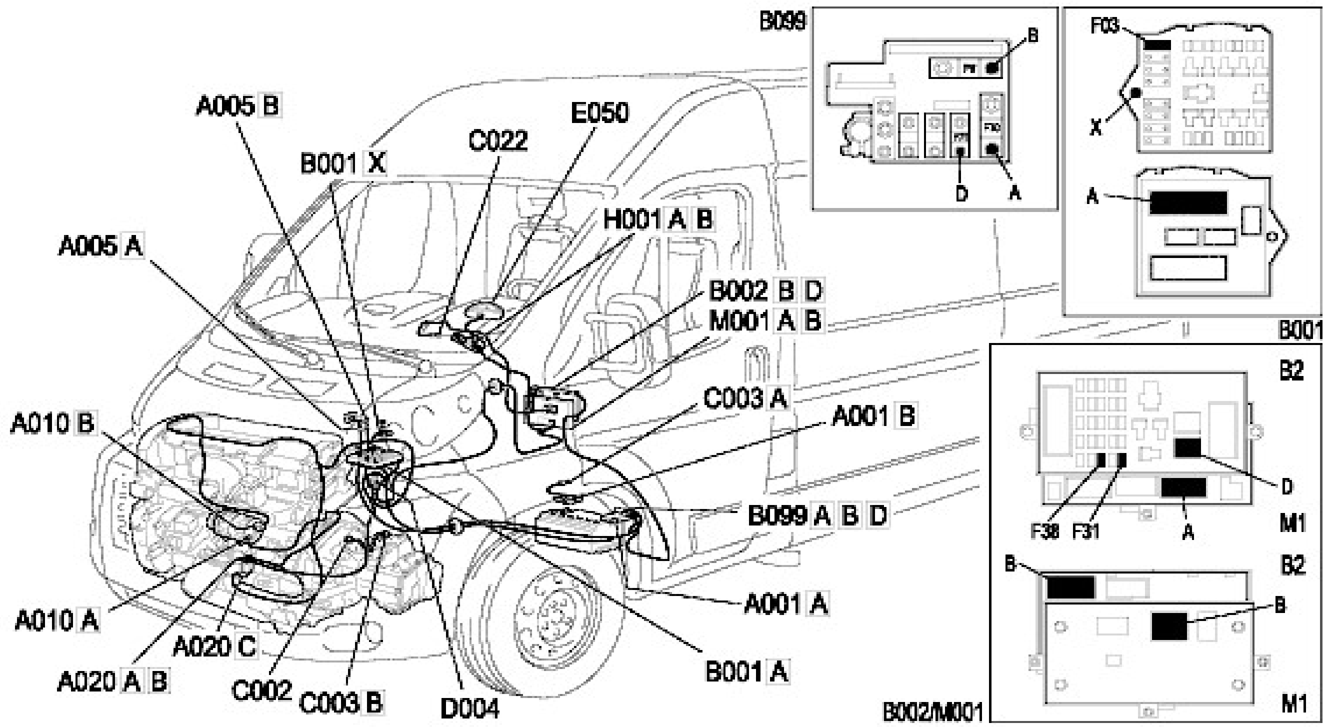
Once a fault has been discovered, the M001 Body Computer, through the CAN, will light up the "battery recharging" warning light on the E050 instrument panel.

STARTING AND RECHARGING - WIRING DIAGRAM



Component code	Description	With reference to the assembly
A001	BATTERY	Op. 5530B BATTERY AND LEADS
A005	Contact board	Op. 5530B BATTERY AND LEADS
A010	ALTERNATOR	Op. 5530A ALTERNATOR AND COMPONENTS
A020	STARTER MOTOR	Op. 5520B STARTER MOTOR AND COMPONENTS
B001	JUNCTION UNIT	Op. 5505A MULTI-FUNCTION COMPONENTS
B002	JUNCTION UNIT UNDER DASHBOARD	Op. 5505A MULTI-FUNCTION COMPONENTS
B099	MAXI FUSE BOX ON BATTERY	Op. 5530B BATTERY AND LEADS
C002	BATTERY EARTH ON ENGINE	-
C003	BATTERY EARTH ON BODYSHELL	-
C022	Centre dashboard earth	-
D004	FRONT/ENGINE COUPLING	-
E050	INSTRUMENT PANEL	Op. 5560B ANALOGUE CONTROL PANEL
H001	IGNITION SWITCH	Op. 5520A IGNITION SWITCH
M001	BODY COMPUTER	Op. 5505A MULTI-FUNCTION COMPONENTS

STARTING AND RECHARGING - COMPONENT LOCATION



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B002	JUNCTION UNIT UNDER DASHBOARD	Op. 5505A MULTI-FUNCTION COMPONENTS
B099	MAXI FUSE BOX ON BATTERY	Op. 5530B BATTERY AND LEADS
C002	BATTERY EARTH ON ENGINE	-
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E050	INSTRUMENT PANEL	Op. 5560B ANALOGUE CONTROL PANEL
H001	IGNITION SWITCH	Op. 5520A IGNITION SWITCH
M001	BODY COMPUTER	Op. 5505A MULTI-FUNCTION COMPONENTS